What Would It Take for Physics to Favor a Theory of Time?

Bridge Premises, Records, and Conditional Constraints

Working manuscript

21 October 2026 — revised provisional argument draft

Abstract

Physical theories constrain temporal metaphysics, but the target of a constraint must be specified. Presentism, eternalism, and accounts of becoming do not, by their labels alone, determine experimental probabilities. This draft argues for a contrast-sensitive burden of proof: an inference from physics to a temporal ontology should identify an admissible physical realization, the bridge premises connecting its structures to existence or passage, and the alternative that the evidence or argument discriminates against. We distinguish physical incompatibility, empirical discrimination, and representation equivalence. Special-relativistic simultaneity and relational quantum clocks illustrate why these inferential forms cannot substitute for one another. In the relativistic case, a definability theorem fixes which becoming relations survive given its premises, and a coupled preferred frame is empirically testable where a stipulated one is not. In the clock case, where relational equivalences fail under interaction depends on how the constraint is formulated. An elementary finite-history example makes record preservation precise while exposing its limits; sequential growth of causal sets supplies a published structural analogue. The position is neither that physics is irrelevant to metaphysics nor that empirical equivalence makes ontologies identical. The proposed framework and examples are expository and argumentative; no mathematical or philosophical priority, new physical prediction, or preferred temporal ontology is established here.

Draft status. Citations vary in reading depth; the companion `SOURCE_AUDIT.md` records those limits. The manuscript does not report a completed systematic review, a proof-assistant verification, or an original result that has passed the project's independent novelty audit.

1 The question is contrastive

What would it mean for physics to favor eternalism over presentism, or local becoming over a moving spotlight? At least three answers are possible. A view might contradict a well-supported physical structure. A realized model might predict observations more successfully than a rival. Or the physical theory might provide an explanation whose ontological commitments are better understood in one way than another. These are serious routes to philosophical justification, but they have different premises and different conclusions.

The danger is not merely an insufficient number of citations. It is an inference in which a result answering one question is taken to answer another: the absence of a distinguished simultaneity relation becomes the nonexistence of passage; the representation of an entire history becomes determinism; or an equivalence of relational dynamics becomes the elimination of fundamental time. A dense catalogue of physical theories cannot repair that inferential gap without identifying the premises that cross it.

Our methodological starting point is physics-constrained but non-reductive. Empirical adequacy matters greatly, while meaningful claims about grounding, existence, and explanation are not

defined away because they lack a direct meter reading. This starting point is itself a philosophical commitment, not a theorem of physics. The breadth of the field is usefully represented by Callender (2011). The contrast between the programs of Callender (2017) and Maudlin (2007) motivates our caution about treating one account of passage as the neutral interpretation of scientific practice; that methodological inference is ours.

Our thesis is therefore a burden-of-proof claim:

> An argument that physics favors a temporal ontology must identify the realization and bridge premises under which the physical result bears on that ontology, and specify the contrast that the result actually supports.

This does not require every philosophical argument to predict a new measurement. It requires an argument to say whether its force is empirical, logical, explanatory, or a combination, and to preserve that scope at its conclusion.

2 What is being compared?

2.1 Independent questions

For this draft, presentism restricts unrestricted concrete existence to what is present; eternalism admits past and future entities as well as present ones. Growing-block views retain the past and present while rejecting future entities; moving-spotlight accounts combine a complete temporal ontology with objectively distinguished presentness. These are schematic families, not complete theories. Direct defenses differ substantially: compare Bourne (2006), Correia and Rosenkranz (2018), and Cameron (2015). Their detailed replies have not yet been fully audited for this manuscript.

Several further questions must remain separate. Are tensed facts irreducible? Does something objectively become present, existent, or definite? Do laws and complete appropriate data select one history? How do objects persist? Is time fundamental? Which asymmetries need explanation? An eternalist can accept indeterministic laws. A four-dimensional spacetime does not by itself make ordinary objects temporal worms. Knowing that an event is now is not automatically knowing an extra objective spotlight property.

These distinctions are not concessions invented to protect a preferred position. McTaggart's argument already differentiates temporal order, change, and A-determinations (McTaggart, 1908); its contested premises cannot be replaced by the phrase "time is an illusion." Likewise, Skow (2015) distinguishes robust from less demanding senses of passage. Clock-rate comparisons constrain metric relations. Connecting those relations to one of these senses of passage requires a further bridge argument.

2.2 Realizations, not bare labels

Represent a comparison target schematically as

$$M = (P, O, B, A, S; \mathcal{E}),$$

where P specifies a physical model and dynamics, O an ontological interpretation, B the bridge premises connecting physical structure to ontological claims, A auxiliary and boundary assumptions, S semantic conventions, and $\mathcal{E}$ the allowed preparations, observations, and interventions. The tuple is bookkeeping rather than a new physical formalism; its components need not be independent.

An ontology may have several proposed realizations. Write $\mathcal{R}(O)$ for the specified admissible class, where admissibility must be defended independently of rescuing a view after each objection.

An empirical likelihood is supplied by a realized package and an experimental protocol, not simply by its label. This observation does not make ontological hypotheses impossible to assess: one may compare independently justified realization classes or marginalize over auxiliaries with explicit priors. It blocks only the unannounced move from a result for one realization to a verdict on every realization.

Constraining admissibility without settling the dispute by fiat. Before inspecting a proposed discriminator, register at least four conditions: (i) mathematical consistency and empirical adequacy in the declared regime; (ii) fidelity to the ontology's explicitly stated constitutive commitments, rather than merely retaining its name; (iii) a specified dynamics, probability model and operational interface, including proposed auxiliaries; and (iv) an independent motivation for additional structure, assessed against the same standards used for rivals. The first conditions constrain which models enter the comparison. Simplicity, explanatory contribution and costs of primitive structure then help rank the survivors; they are not covert empirical vetoes.

The proponent bears the burden of describing a realization and defending its bridges; the exclusion advocate bears the stronger burden of defending the scope of the excluded class. Neither party may change those standards silently after a counterexample. Where reasonable standards disagree, report conditional verdicts under both rather than call one class uniquely admissible. This does not mechanically settle abduction, but it makes auxiliary rescue and asymmetric demands on rivals inspectable. The constitutive commitments and permitted auxiliary changes should be recorded before evaluating the discriminator. Symmetric treatment alone does not show that an admissibility criterion is independent of the intended verdict. A comparison should therefore state which conclusions change under reasonable alternative admissibility profiles. Defending those profiles remains a task for the stronger philosophical argument, rather than a result of this checklist.

A worked case: Brading's law-constituted present. Brading (2013, 2015) accepts special relativity as a complete account of spatiotemporal structure in its regime, but not as a complete theory of matter dynamics. She proposes that what exists presently is fixed by the extent of a dynamical system's law-constituted unity rather than by a simultaneity hyperplane. To avoid the block, that extent must be smaller than the universe's full history; it need not be smaller than the system's own complete history. The extent is system-relative and need not be transitive, so the view does not imply that everything outside a chosen system's present is unreal. Brading herself leaves open whether there are genuine dynamical subsystems smaller than all of spacetime, and she supplies no common solution, access protocol or probabilities against which a rival could be scored (Brading, 2015, Sec. 5, p. 18).

The strongest block-compatible rival in this case is a composite we assemble, not a position any single cited author develops against Brading. It grants law-based dynamical unity but denies that a system's law-sustaining region exhausts that system's existence. Explanatory arguments about the direction of explanation between spacetime structure and dynamics bear on it: Balashov and Janssen (2003) argue that spacetime structure unifies the Lorentz invariance of matter laws, against a neo-Lorentzian rival (Secs. 9–11 of the longer PhilSci archive preprint, the version read), and Brown and Pooley (2006) defend the reverse direction (Secs. 3–5 of arXiv:physics/0403088v1). Neither addresses Brading's ontology directly. If both realizations are granted a common, consistent and empirically adequate model, matched likelihoods give no empirical preference, and in the cited texts no independent reason decides between the restricted-existence bridge and the full-history bridge. The outcome is a conditional underdetermination, which is what the framework predicts when bridges differ and predictions do not. Any matched-likelihood illustration is our stipulation,

not a property either source establishes.

3 Three different inferential burdens

3.1 Exclusion

A physical incompatibility argument has a form such as

$$O \wedge B \wedge A \implies F, \qquad P \implies \neg F.$$

It follows that P excludes the conjunction $O \wedge B \wedge A$. Concluding $\neg O$ requires independent support for the indispensable bridge and auxiliary premises, or an argument that every admissible realization of O encounters the conflict. Otherwise the result identifies a conditional cost: retain the ontology only by revising the disputed structure or bridge.

This is not the claim that a bare ontology can always be rescued. A bridge may be constitutive of a precisely defined view, or its removal may destroy the explanation that made the view attractive. That stronger argument should be made rather than hidden in a generic ontology label.

3.2 Empirical discrimination

Empirical discrimination requires different probability assignments to a common experimental question. At fixed realized packages, if

$$p(D \mid M_1, e) = p(D \mid M_2, e)$$

for every outcome event D and protocol $e \in \mathcal{E}$, those observations cannot discriminate the packages by their likelihoods. This statement is restricted to the declared protocol class, not all conceivable future observations.

For composite hypotheses, pointwise equivalence at matched parameter values is insufficient for equal Bayesian evidence. The marginalized likelihoods must also agree. For example,

$$p(D \mid O_i, e) = \int p(D \mid \theta_i, O_i, e)\, \pi_i(d\theta_i \mid O_i, e)$$

can differ because the realization or parameter priors differ. Equal prior-predictive distributions give a Bayes factor of one wherever the common likelihood is nonzero; equal predictions at selected parameter settings need not. Prior preference remains possible, but must not be redescribed as an empirical likelihood advantage.

3.3 Representation equivalence

A structure- or probability-preserving map can establish equivalence for its specified domain. It does not automatically establish identical metaphysical interpretations, gauge redundancy, or equal explanatory value. Nor does a mathematically invertible reparameterization necessarily preserve which operations an observer can perform. An equivalence argument must state the common state space, physical inner product when relevant, observable map, and access restrictions. A failure of such an equivalence can be physically important without immediately deciding an ontology.

4 Case I: what a relativistic present would require

The elementary simultaneity result is exact within its standard domain. A boost along one spatial axis gives

$$t' = \gamma(t - vx/c^2), \qquad \gamma = (1 - v^2/c^2)^{-1/2}, \qquad |v| < c.$$

An event $(0, L)$ with $L \neq 0$ is simultaneous with the origin in one frame but has $t' = -\gamma v L/c^2 \neq 0$ for a nonzero boost. Thus that simultaneity hyperplane is not invariant under all boosts fixing the origin. This is standard kinematics, not a new result or a proof that nothing changes.

To obtain an ontological conclusion one must say why presentness must be that kind of hyperplane and respect the relevant symmetries. Putnam's argument invokes an observer-neutral reality-transfer principle; its broader closure argument includes further assumptions about observers (Putnam, 1967).

The definability literature shows how much those premises carry. Clifton and Hogarth (1995, Thm. 1, p. 365) prove that a worldline-independent becoming relation in time-oriented Minkowski spacetime that is implicitly definable from the time-oriented metrical relations, transitive, and contains the chronological past can only be past chronological connectibility, past causal connectibility, or the universal relation. "Definable" there means invariance under time-orientation-preserving automorphisms (p. 361). The premises are not shown to be individually necessary, but two are tested: without time orientation only the universal relation survives (Cor. 1), and once privileged worldlines are allowed, families of simultaneity planes and past null cones satisfy the remaining premises (Sec. 3, pp. 376–380). The theorem is therefore a precise exclusion of the conjunction of its premises. Which of them a temporal ontology must accept is the bridge question; the invariance reading of "objective" does most of the work. Stein's event-relative approach supplies a different account of definite past structure (Stein, 1991). The present draft does not reproduce Stein's characterization theorem: the isolated OCR implication is false as stated without additional contextual constraints. The surrounding argument includes causal-past inclusion and unsettled states; comparison with the original printed theorem remains unresolved. No error in the printed paper is inferred.

One response adds a preferred foliation. That is a substantive extra structure, not a free conclusion of SR. But observability also requires care. Appending a structure without changing old dynamics, probability assignments, or accessible operations need not change the old predictions. This conditional observation does not establish that any particular preferred-foliation theory is admissible or empirically equivalent; a coupling can make the structure observable. Nor does mere covariance of a notation establish absence of a physical preferred structure.

The difference is concrete. In Einstein-aether theory a dynamical unit timelike vector couples to gravity, and its preferred-frame effects appear in post-Newtonian parameters such as α_1 (Foster and Jacobson, 2006). Laser ranging of the LAGEOS satellites over about three decades gives $\alpha_1 = (+1 \pm 2) \times 10^{-5}$ and $(+2 \pm 3) \times 10^{-5}$ from two orbit-determination codes, with a systematic error near 8×10^{-6}, compatible with zero (Lucchesi et al., 2026). That result constrains a coupled preferred-frame package. It does not test a foliation added to unchanged Lorentz-covariant dynamics with no coupling, which is empirically equivalent to special relativity by construction for those observables. Neither outcome selects presentism over eternalism; each locates which package an observation can reach.

Another response disputes that simultaneity must provide the unity of what exists. Brading develops such a presentist program (Brading, 2013, 2015), worked through in Section 2: it relocates the contested premise from simultaneity to the extent of law-constituted unity, and it supplies no quantitative discriminator. That is enough to identify a premise needing argument, not enough to vindicate the whole program.

There are real philosophical costs on both sides. A preferred structure may be explanatorily idle; a local becoming relation may do no more than restate causal order; and a block account owes an account of the phenomena motivating passage. The point is to locate these costs accurately. A geometrical constraint on one global-present realization is not a completed comparison of all these packages.

The strongest schematic foliation reply is not that the structure must be detectable. It is that a primitive foliation constitutes which entities exist together, even if it changes no measured quantity. A critic can question its motivation or its explanatory return; merely noting observational redundancy does not answer a constitutive claim. The same standard applies to a local becoming bridge: it could be defended as specifying definiteness rather than causing a new detector response, but must explain what motivates that role instead of simply assigning a temporal name to the causal past. These are deliberately schematic rival arguments, not fully audited reconstructions of a particular defender. The comparative explanatory judgment remains open.

5 Case II: relational dynamics without an ontological shortcut

Relational clock constructions expose a second inference risk. A constrained global description can admit conditional evolution relative to a chosen clock. The relational-trinity work relates Dirac-observable, conditional Schrodinger, and reduced Heisenberg descriptions under specified hypotheses (Höhn et al., 2021). Its noninteracting setup and clock conditions matter. Interaction treatments already exist, including the kernel equation studied by Smith and Ahmadi (2019); "add interactions" is not a generic unoccupied research gap.

Two implications should be resisted. First, a stationary constraint equation does not by itself mean that subsystems cannot change: correlations can support conditional dynamics. Second, successful conditional dynamics does not by itself establish that correlations exhaust temporal reality. A reductionist argument needs a defensible sufficiency principle identifying all relevant temporal roles and showing that the construction realizes them. A primitive passage advocate cannot dismiss the construction, but may dispute that exhaustiveness premise.

The inverse mistake is equally important. An interaction-dependent or nonunitary conditioned evolution does not establish objective passage or an open future. One must specify the physical state space, clock instrument, normalization, accessible measurements, and what the proposed ontology would predict differently. A transformation that changes access restrictions cannot be treated as experimentally neutral merely because it is invertible on an enlarged mathematical space.

Where does a published equivalence stop? The trinity results assume that the clock and system do not interact (Höhn et al., 2021), and the periodic-clock extension makes the same assumption, noting that interactions would typically spoil the clock's periodicity (Chataignier et al., 2026). With interactions, whether clock-conditioned states stay normalized, and so whether a reduced unitary description exists, depends on the model and on how the constraint is written. For one class of constraints the normalization is constant in clock time only for interactions of a commuting type (Cafasso et al., 2024). Yet there are finite examples with interactions that do not commute with the clock Hamiltonian and still keep unit norm (Smith and Ahmadi, 2019; Boette and Rossignoli, 2018; Ludescher et al., 2026), and a discrete-time formulation recovers unitary conditional evolution for general clock-controlled interactions (Hausmann et al., 2025, Sec. 5).

A worked finite check of our own makes the dependence explicit; it is an application of these constructions, not a new result. With a three-state clock, a qubit and an interaction that fails to commute with the clock Hamiltonian, the Dirac and conditional expectation values coincide identically on physical states, because physical states are exact eigenvectors. The descriptions part

only at normalization: reading probabilities become non-uniform and the reduction maps cease to be isometries, while a commuting control stays unitary. If the constraint is an eigenvalue equation for the full Hamiltonian, a generic interaction leaves one-dimensional physical spaces, so there is no family of initial system states and no dynamical map whose memory could be assessed. If the constraint is invariance under a clock step or a group action, the physical space survives. The philosophical consequence is a bridge point: whether "relational time" is equivalent to an external-time description under interaction is not settled by the physics until the constraint formulation is fixed, and that choice is itself a premise.

6 A finite record example and its exact limits

The following elementary construction makes one possible equivalence burden explicit. It is an expository application of ordinary probability, not a novel theorem about becoming.

Proposition 1 (Probability preservation under matched kernels)**.** *Let (X, μ) and (Y, ν) be probability spaces, and let $F : X \to Y$ be measurable with $F_*\mu = \nu$. The outcome kernels share one measurable outcome space. For each allowed complete observation protocol e, suppose $K_X^e(x, D) = K_Y^e(F(x), D)$ for μ-almost every x, for every outcome event D. Then the two packages assign the same probabilities to those protocols.*

Proof. For any declared e, D, the probability in X is

$$\int_X K_X^e(x, D)\, d\mu(x) = \int_X K_Y^e(F(x), D)\, d\mu(x) = \int_Y K_Y^e(y, D)\, d\nu(y),$$

which is the probability in Y. The result assumes rather than derives the measure and kernel preservation. □

For a finite horizon T and finite state space S, let

$$\mu(x_0, \ldots, x_T) = \pi(x_0) \prod_{t=0}^{T-1} P_t(x_{t+1} \mid x_t).$$

One implementation samples an initial state and then successive transitions; another samples an entire path with this law and releases its entries in order. Any common protocol restricted to the revealed prefixes, whose auxiliary randomness has the same conditional law given those prefixes and no access to the unrevealed path, yields the same record distribution. Adaptive choices of which revealed data to read can be included in the complete kernel. Operations modifying future transitions are not included unless their conditional dynamics are separately matched.

An exact numerical illustration. Take $S = \{0, 1\}$, $T = 2$, a fair initial state, and at each step a probability $3/4$ of staying in the same state. The path 000 has probability $(1/2)(3/4)(3/4) = 9/32$; 010 has probability $1/32$. Both implementations assign these same probabilities and the same prefix marginals. A companion standard-library check enumerates all paths using exact fractions. It verifies this example, not the general measure-theoretic argument or a theory of time.

Several limits are decisive. A sequential generator is not thereby the growing block theory; a sampled path is not thereby eternalism. The observer-access condition, the measure, and the identification of records do the operational work. If a different package gives access to unrevealed entries, or responds differently to interventions, the stated equivalence need not apply. Such extra access is a changed model assumption, not something entailed by one ontology label.

Observational agreement also falls short of interventional agreement. Consider a fair latent bit Z. In model A set $X = Z$ and $Y = X$; in model B set $X = Z$ and $Y = Z$. Both observational distributions have $X = Y$ with a fair marginal. Here Z is exogenous in both models. Under an intervention replacing only the assignment for X by $X := 0$, model A has $Y = 0$, whereas model B retains a fair Y. This familiar type of causal example shows why accessible interventions must be specified. It does not attribute either causal model to a temporal ontology.

The philosophical payoff is limited but useful: records cannot favor one implementation when measure and complete allowed response kernels are matched. To turn that into an argument about becoming requires a separately defended interpretation. Calling the implementations two ontologies would skip exactly the premise the paper asks us to inspect.

Sequential growth of causal sets supplies a published, nontrivial analogue. In classical sequential growth, a causal set is built by adding one element at a time, and the dynamics is required to be generally covariant (Rideout and Sorkin, 1999): conditional on the unlabelled causal set at a given stage, every natural labelling, that is every birth order compatible with it, is equally likely (Brightwell and Luczak, 2011). Birth order is then ancillary: the data a covariant observer can access carry no information about the order of growth, just as the revealed prefixes above carry none about how the path was generated.

The growth picture also meets the completion question directly. No nontrivial event decided at a finite stage of growth is covariant (Surya and Zalel, 2020), so the gauge-invariant questions concern completed histories. Several authors have read this as making a growing-block construal of classical growth depend on the block it was meant to replace (Sorkin, 2007; Arageorgis, 2016; Wüthrich and Callender, 2017). In complex (quantum-like) generalizations the tension sharpens: for amplitude parameters outside $[0, 1]$ the stage-wise amplitudes are well defined but admit no countably additive extension to the covariant events (Dowker et al., 2010; Surya and Zalel, 2020). Read with care, this excludes a conjunction on each side. A growing-block reading that takes only finitely settled, birth-order-independent facts as real has no nontrivial facts, for any dynamics. A block reading that demands a countably additive measure on completed histories has none for these parameter values. Neither result discriminates between the ontologies without those bridge premises, and the cheapest revision may be to the model or to countable additivity rather than to either ontology.

7 Objections and replies

Bridge premises are unavoidable; your test is too demanding. Correct: every interpretation uses bridges. Their presence is not a defect. The demand is that the bridge be visible, defensible, and relevant to the contrast. A bridge can be constitutive of a view or supported by explanatory arguments. We reject the inference "extra premise, therefore arbitrary," as firmly as "physical equation, therefore metaphysical conclusion."

This makes every view compatible with physics. Only if admissibility is unconstrained, which the framework rejects. A view that discards its defining commitments, adds ad hoc structure, or fails empirical adequacy is not rescued by renaming the remainder. Conversely, showing one admissible realization establishes compatibility, not equal plausibility. A full exclusion claim has a stronger quantifier burden than a counterexample to one proposed proof of exclusion.

Relativity and parsimony already favor the block. They may favor a specified block package over a competitor with unjustified structure. The paper's demand is to articulate that explanatory comparison: which structure is avoided, whether the rival needs it, and what explanatory work

competing packages perform. An inference to the best explanation is not discredited by being philosophical rather than a direct detector reading. It is also not rendered deductively compulsory by the success of SR.

Phenomenology supplies evidence for passage. It supplies phenomena a satisfactory account should explain. It does not automatically choose among explanations. The advocate of primitive passage must show why experience supports that cause over alternatives; the critic must explain the experience rather than simply call it illusory. The accounts discussed by Callender (2017) and Skow (2015) are relevant comparators, not settled answers inherited by this draft.

An inaccessible ontology is physically meaningless. That conclusion needs a verificationist or other semantic argument which we have not assumed. Empirical equivalence can coexist with differences in grounding, explanation, simplicity, or modal commitment. The narrower conclusion is that matched observational likelihoods do not themselves select between the packages. Physics can still contribute through structural constraints and explanatory considerations.

Your record example is trivial and deliberately restrictive. The mathematics is elementary and the restrictions are deliberate. It is not offered as the novel contribution. Its purpose is to distinguish a proved conditional equivalence from an unsupported claim about all ontologies. A publishable extension would need a nontrivial physical class, a justified access model, and a result not already provided by the nearest predecessors.

8 What would turn this into a research contribution?

The present argument supplies a way of auditing claims, not a discovery that underdetermination exists. At least three developments could make it more than an organized restatement of familiar cautions:

1. A source-faithful reconstruction showing that a widely used inference requires a neglected bridge, and that the strongest published defenses and replies do not already establish the same point.

2. A nontrivial classification of when specified physical realizations preserve records and interventions, with explicit counterexamples at the boundary and a comparison to known equivalence results.

3. A substantive defense of one bridge principle that survives alternative temporal interpretations and explains why the resulting constraint is not merely a preference built into a definition.

The independent novelty gate is unmet. Neither the elementary proposition nor the use of agents, Python, or a future Lean proof supplies priority. Source audits, model reproductions, and the strongest contrary philosophical arguments are still needed. General relativity and cosmological time also require a separate extension: the existence of smooth Cauchy slicings under global hyperbolicity (Bernal and Sánchez, 2005) is not uniqueness of a metaphysical present.

9 Provisional conclusion

Physics can make temporal metaphysics harder, more precise, and sometimes empirically vulnerable. It does not follow that every temporal distinction is physically idle; nor does a successful mathematical

representation settle all questions of existence and passage. The appropriate unit of comparison is a defended ontology–physics–bridge package with explicit access conditions. An argument should state whether it excludes such a package, discriminates its predictions, or establishes a representation equivalence. Keeping those conclusions distinct is a precondition for discovering, rather than assuming, what physics tells us about time.

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